Michael Theologitis

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PERSONAL STATEMENT

I study how to build AI systems that manage, reuse, and self-organize their accumulating context to solve long-horizon, data-intensive tasks. My work spans context engineering, memory, and agent harnesses, extending to recursive self-improvement. My broader goal is to help journalists, lawyers, and scientists answer complex questions over large collections of heterogeneous data in ways that are trustworthy and inspectable.

EDUCATION

UNIVERSITY OF WASHINGTON | *PhD in Computer Science* 2025 – Present
- Advisors: Dan Suciu, Yulia Tsvetkov
- Interests: NLP, AI Agents, Data Management

TU-CRETE | *MS in Computer Science and Engineering* 2023 – 2025
- Supervised Fine-Tuning, LLMs, Federated Learning

TU-CRETE | *BEng & MEng in Electrical and Computer Engineering* 2017 – 2023
- Deep Learning, Convex Optimization, Linear Algebra, Statistics

EXPERIENCE

CREXDATA Europe
Research Software Engineer 2023 – 2025
- Developed a novel algorithm, reducing costs by 10×–100× for Federated Learning jobs.
- Built a platform to deploy and monitor 200+ concurrent FL jobs, each with 1,000+ devices on HPC environments (e.g., Barcelona's MareNostrum supercomputer).
- Fine-tuned LMs on uncertain streaming data to monitor and alert first-responders for emergency scenarios (e.g., floods, fires, etc.) for the Ministry of Interior of Finland.

SELECT PUBLICATIONS

Preprint ***Michael Theologitis****, ..., Benjamin Newman, Yulia Tsvetkov, Dan Suciu* Realize What Matters: Principled Context Representation for Large-Scale Reasoning

NeurIPS '26 *Dean Light**, ***Michael Theologitis****, ..., Pang Wei Koh, Dan Suciu, Yulia Tsvetkov* Deep Reasoning in General Purpose Agents via Structured Meta-Cognition

ACL '26 ★ Top 15% ***Michael Theologitis****, Preetam Dammu, Chirag Shah, Dan Suciu* ClaimDB: A Fact Verification Benchmark over Large Structured Data

AAAI '26 ★ Oral ***Michael Theologitis****, Dan Suciu* Thucy: An LLM-based Multi-Agent System for Claim Verification across Relational Databases *LaMAS'26 Workshop*

CIKM '26 ***Michael Theologitis****, Vasilis Samoladas, Antonios Deligiannakis* FDA-Opt: Communication-Efficient Federated Fine-Tuning of Language Models

***** Equal contribution

TECHNICAL SKILLS

Programming Python (10+ years), Claude Code, SQL (5+ years), C, Java, MATLAB

Agentic Frameworks OpenAI Agents SDK, Smolagents, PydanticAI, Claude Agent SDK

AI/ML HuggingFace, vLLM, PyTorch, TensorFlow, NumPy, Ollama, Pandas, Matplotlib, Seaborn

Databases PostgreSQL, MySQL, SQLite, MongoDB

Parallel Computing Spark, Dask, Hadoop, Kafka, HDFS, Yarn, Slurm, Docker

Tools & Platforms Linux, GitHub, Git, PyCharm, VS Code, JupyterLab, LaTeX, Powerpoint

SELECT PROJECTS

R^{3}CON: EFFECTIVE CONTEXT REPRESENTATION | *Link* Jan. 2026 – Present

- Developed R^{3}CON, a harness that helps LLMs **piece together interdependent evidence** scattered across large document corpora to solve tasks beyond their context limits.
- Translated the cognitive theory of relevance realization into **design principles for building AI systems that effectively organize large contexts for reasoning**; analyzed how existing systems' successes and failures relate to these principles.
- Evaluated against 9 state-of-the-art AI systems, improving accuracy over the strongest baseline by **20 and 8.4 percentage points** on CorpusQA and Loong, respectively.
- R^{3}CON enables smaller models to outperform much larger ones, with 4B and 9B models surpassing all evaluated 35B baselines on Loong and CorpusQA, respectively. Moreover, Qwen3.5-35B-A3B with **R^{3}CON outperforms Claude Code equipped with Claude-Sonnet-5** on CorpusQA at **3.7× lower cost**.

INSTILLING HUMAN META-REASONING IN AGENTS | *Link* Nov. 2025 – Feb. 2026

- Built a prompting strategy to take **human meta-reasoning** and convert it **into in-context examples** that agents can understand and deploy.
- Instilled the flexibility of human reasoning in general-purpose agents.
- Introduced the **DOLORES** agent and evaluated it across four hard benchmarks (e.g., long-horizon, multi-hop, deep search) improving over SOTA by **+24.8% accuracy** on average.

CLAIMDB: BENCHMARK FOR AGENTS | *Project Site* Nov. 2025 – Feb. 2026

- Benchmark for fact-verification over large relational databases: **80 DBs** and **53k facts**
- **For each fact**, it requires that agents explore structured data that is **+110M tokens** large
- Evaluated **30 state-of-the-art LLMs**, and showed models struggle in reasoning over large structured data: they **struggle in *abstention*** — the ability to admit *"I don't know"*.

THUCY: MULTI-AGENT SYSTEM FOR FACT VERIFICATION | *Link* Sept. 2025 – Nov. 2025

- Explores databases to uncover whether *facts* are supported from the structured data.
- Improved over the SOTA on the TabFact benchmark by **+5.6 percentage points**.

FDA-OPT: A FEDERATED OPTIMIZATION ALGORITHM | *Link* Sept. 2024 – Aug. 2025

- FDA-Opt completely eliminated the hard-to-calibrate communication interval τ from the family of optimization algorithms FedOpt (e.g., FedAvg, FedAdam, etc.).
- Fine-tuned LMs (encoders) on various NLP tasks and found that **FDA-Opt converges better than FedOpt**—across all individual optimizers like Adam, AdamW, SGD, SGDM, etc.—and is also around **2× more efficient**.

INVITED TALKS

RelationalAI Fact Verification over Structured Data: Agentic Systems and Benchmarks 2026

COMMUNICATION

Languages English (Proficient), French (Beginner), Greek (Native)

Presenting Delivered 50+ presentations of highly technical material to audiences of 40+ academic and industry partners, with a strong track record of creating clear, engaging, and effective slides for expert and non-expert audiences alike

Writing Strong academic and research writing skills, demonstrated through multiple first-author publications